

Module 10: Epigenetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. How does epigenetic regulation differ from a mutation?

- A. Both permanently change the DNA sequence.
- B. Epigenetics deletes entire genes.
- C. Epigenetics changes gene activity without changing the DNA sequence.
- D. Mutations never affect any trait.
- Answer: C
- Why: Epigenetic marks such as methylation change how genes are expressed, not the underlying sequence. A mutation changes the sequence itself.

2. DNA methylation is usually associated with:

- A. reduced transcription of a gene
- B. faster DNA replication
- C. deletion of a chromosome
- D. building proteins directly
- Answer: A
- Why: Methylation typically silences or reduces transcription by making the DNA less accessible.

3. When chromatin is packed tightly, genes in that region are generally:

- A. copied much faster
- B. more likely to mutate
- C. more accessible to be read
- D. less accessible and less expressed
- Answer: D
- Why: Compact chromatin blocks access to genes, so they are expressed less. Open chromatin allows transcription.

4. Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

- A. their DNA sequences must actually be different
- B. environment and epigenetic changes altered their gene expression
- C. one twin is not truly human
- D. epigenetic change is impossible
- Answer: B
- Why: Identical DNA can still show different expression because environment-driven epigenetic changes turn genes up or down.